Exploratory Data Analysis (EDA)

Diabetes Dataset

Author: Vijiyalakshmi Rajeshkumar

Tools Used: Python, Pandas, NumPy, Matplotlib, Seaborn

Dataset: Diabetes Dataset

---

1. Introduction

This report presents an analysis of the Diabetes dataset with the objective of exploring relationships between various physiological attributes and diabetes progression. Understanding these relationships can provide insights into which factors are most associated with the progression of the disease over time. The dataset contains measurements such as Body Mass Index (BMI), blood pressure, and blood serum indicators for a group of patients.

2. Feature Description

The dataset contains the following features:

- **Age:** Age of the patient
- **Sex:** Gender of the patient
- **BMI:** Body Mass Index (measure of body fat)
- **BP:** Average blood pressure
- **S1-S6:** Blood serum measurements (various biochemical indicators)
- **Progression:** Diabetes disease progression after one year

Each feature provides insight into the physiological state of the patient and its potential influence on diabetes progression.

3. Data Cleaning and Missing Data Handling

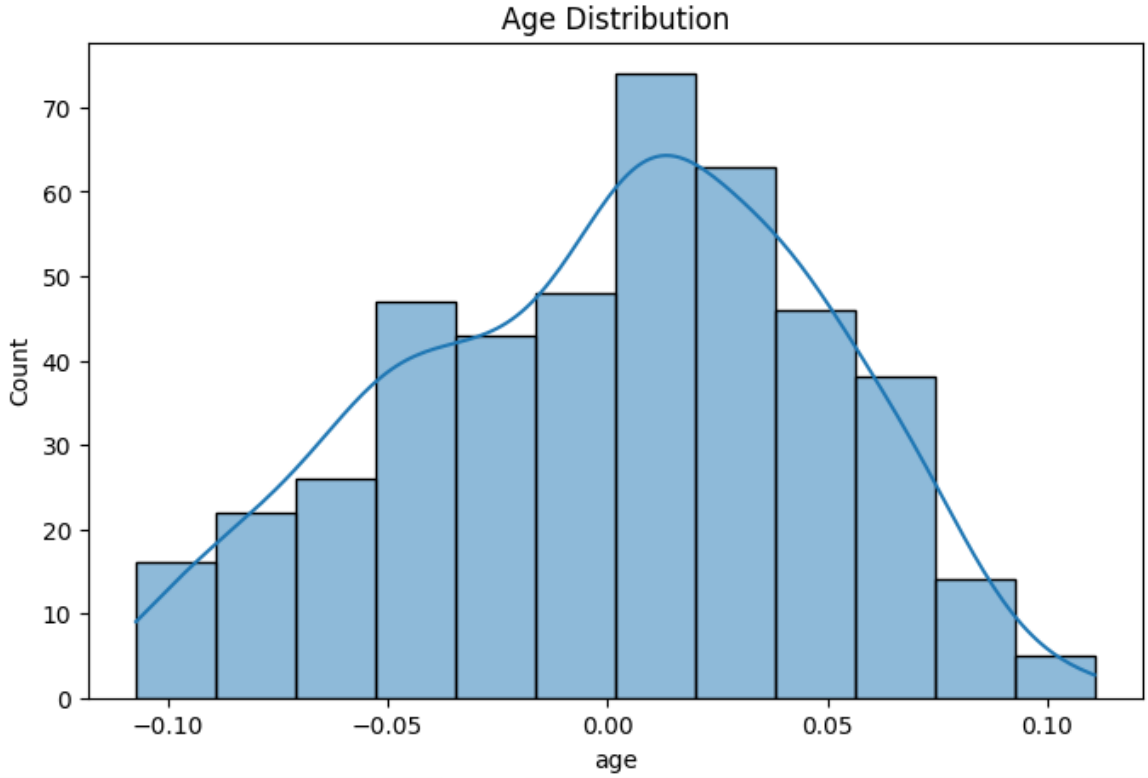
The dataset underwent the following cleaning procedures:

- Column names were standardized by converting them to lowercase and removing any extra spaces.
- Certain variables, such as BMI and blood pressure, contained zero values which are not medically realistic. These values were treated as missing and replaced with the median of the respective columns to maintain data integrity.
- No explicit null values were found in the dataset; however, the treatment of unrealistic zero values ensured that all analyses reflect true physiological trends.

4. Exploratory Visualizations and Observations

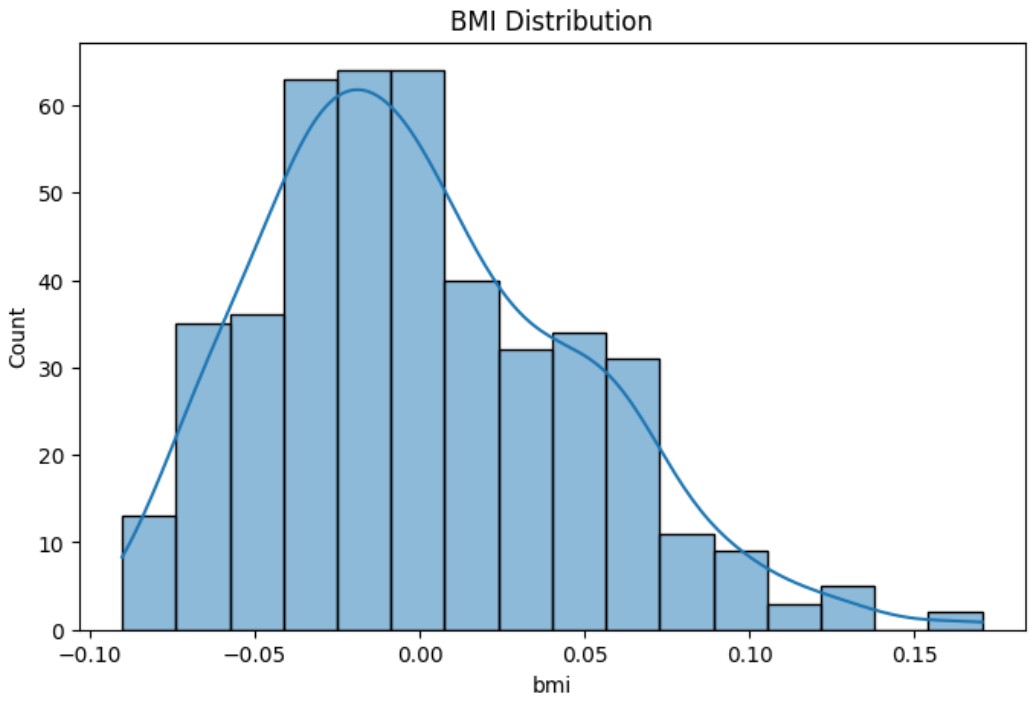
➢ Age Distribution

The age distribution shows that most patients are between 40 and 60 years, indicating that the dataset primarily represents middle-aged and older adults.



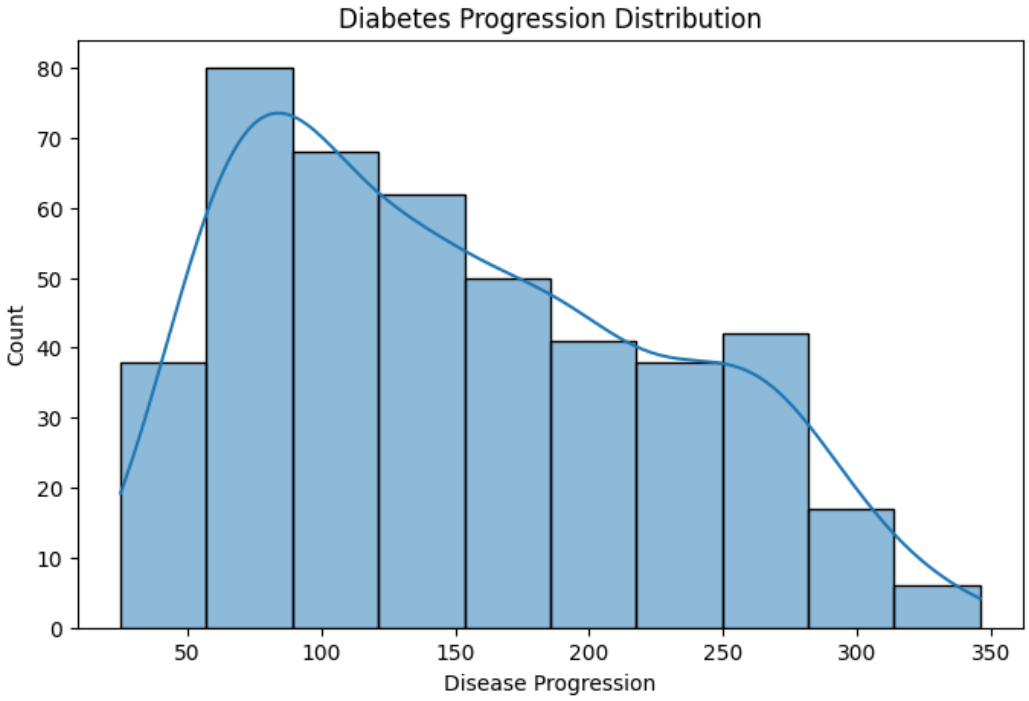
➢ BMI Distribution

BMI values are roughly normally distributed, with most observations clustered near the mean. Extreme BMI values are rare, suggesting most patients fall within a moderate body fat range.



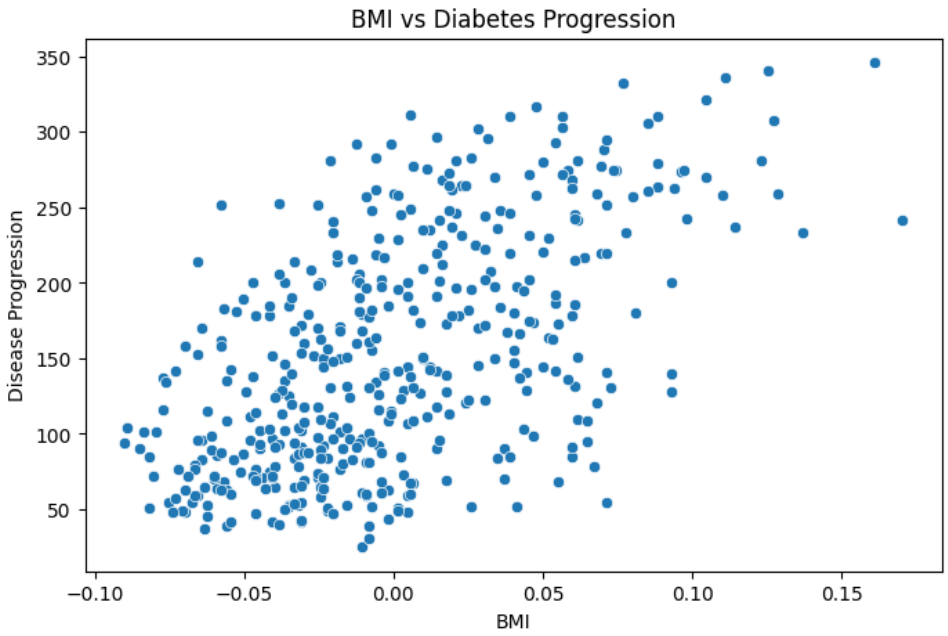
- **Diabetes Progression Distribution**

Diabetes progression values are slightly right-skewed. Most patients have progression values between 50 and 200, while values above 250 (indicating severe disease progression) are less frequent.



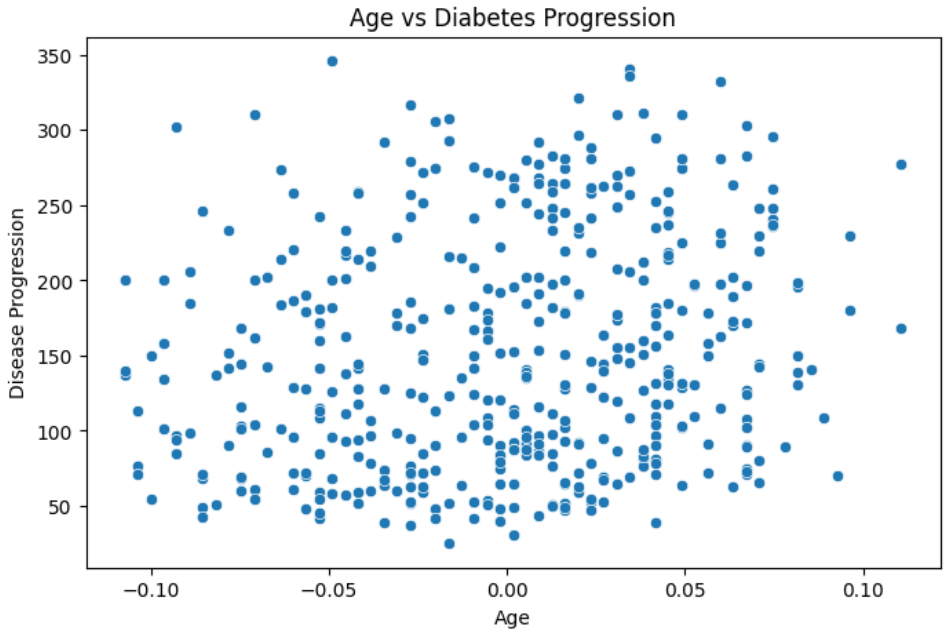
- **BMI vs Diabetes Progression**

The scatter plot suggests a mild positive relationship between BMI and diabetes progression. Higher BMI values are sometimes associated with higher progression scores, but the wide scatter indicates BMI alone is not a strong predictor.



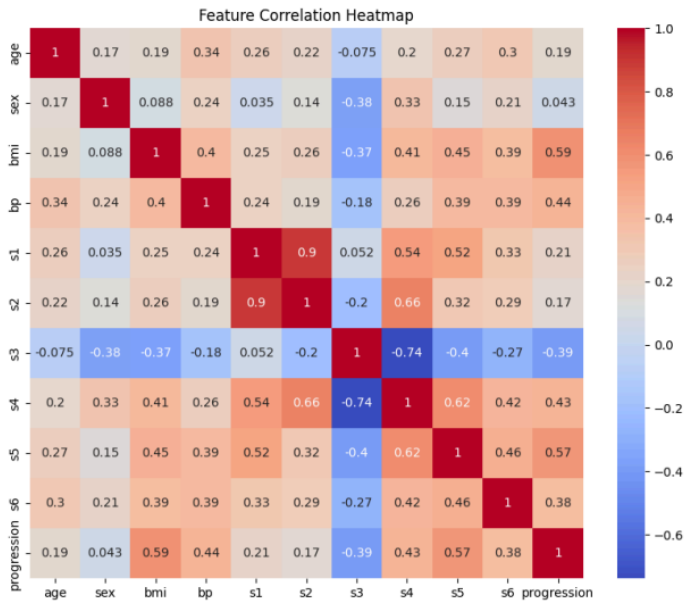
- **Age vs Diabetes Progression**

No clear trend is observed between age and diabetes progression, indicating that age alone may not be a strong predictor.



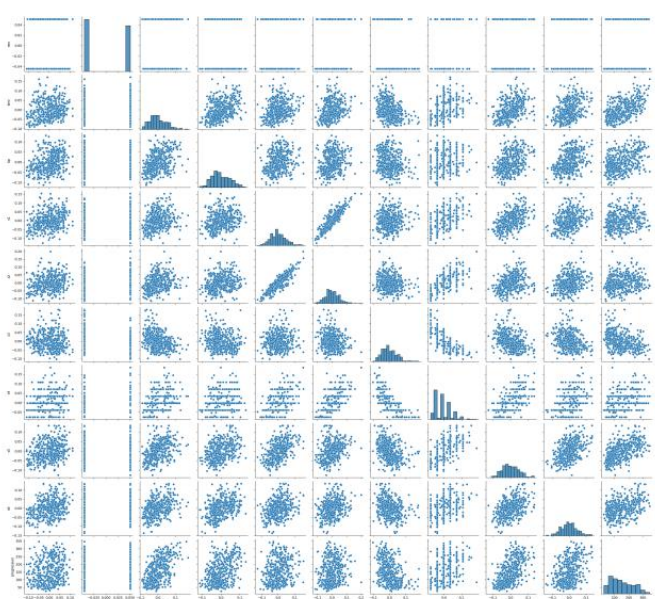
➢ Correlation Heatmap

A correlation analysis reveals that BMI and s5 (a blood serum measurement) have the strongest positive correlation with diabetes progression. Blood pressure (BP) shows a moderate correlation, while age and sex show weak correlations.



➢ Pairplot

The pairplot illustrates pairwise relationships among all numerical variables. It reinforces that BMI and s5 are the most strongly associated features with progression, while age and sex exhibit weak relationships.



5. Key Findings

- BMI is strongly associated with diabetes progression.
- s5 (serum measurement) shows a strong correlation with disease progression.
- Blood pressure (BP) has a moderate positive correlation.
- Age and sex have relatively weak associations with progression compared to other features.

These findings highlight that metabolic and physiological factors play a more important role in disease progression than demographic factors.

6. Conclusion

The exploratory data analysis of the Diabetes dataset indicates that BMI, s5, and blood pressure are the most significant variables associated with diabetes progression. Age and sex, while informative, are less predictive of disease outcomes.

These insights can guide the development of predictive models for diabetes progression and help focus medical monitoring on the most influential physiological factors. Future work could include predictive modeling, feature engineering, or deeper analysis of serum indicators to better understand the disease dynamics.

❖ This report was written by **VIJIYALAKSHMI RAJESHKUMAR**